

WR-V

Experiment and Realization Selection: Branch, Contextualize, Select, or Remain a Class

Mathematical draft v0.1

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Abstract

A new experimental outcome restricts compatible descriptions, but its distribution need not identify how a declared state changes. WR-V adds an experimental transition layer to the response and completion interfaces of WR-I–IV. In a finite classical model, a stochastic instrument specifies an outcome and a successor label jointly. Outcome likelihoods identify only its marginal; the unobserved transition remains a product of simplices. Calibrated successor probes turn this ambiguity into a nonnegative linear inverse problem. We give elementary proofs of a uniform rank criterion, a support-sensitive pointwise uniqueness criterion and a criterion for identifiable linear functionals. A three-state benchmark distinguishes exact boundary uniqueness from ambiguity at interior data and from stability at finite resolution. A contextual parity example and a sequential positive-likelihood lemma delimit two additional uses of the programme’s four scenario labels. Those labels are not an exhaustive partition or competing ontologies inferred from one outcome. The results are finite-model constructions and standard linear/probabilistic arguments, with no theorem-priority claim or physical branching conclusion.

Keywords: experimental update; stochastic instrument; successor-state identification; nonnegative inverse problem; contextuality; exact selection.

Parallel English and Russian texts share all equations and result numbers. REVIEWABLE_DRAFT; not externally peer reviewed; no WR-V DOI. Content: CC BY 4.0. Code: MIT.

Contents

1	Question, interface and claim ceiling	3
2	Global compatibility and finite experimental labels	3
3	Four scenario labels and their different domains	4
4	What outcome likelihoods leave undetermined	4
5	Successor probes and transition-completion certificates	5
6	A three-state experiment and finite resolution	6
6.1	Exact uniqueness and uncertainty regions	6
7	Contextualization changes an identification obligation	7
8	Sequential evidence and exact selection	7
9	Provenance, failure criteria and downstream handoff	8

1 Question, interface and claim ceiling

The founding WREH programme proposes four realization scenarios: BRANCH, CONTEXTUALIZE, SELECT and REMAIN A CLASS. Their mathematical types differ. Selecting a compatible hypothesis is an inference operation. An experimental transition is part of a dynamical model. A family of context-indexed descriptions concerns how variables are identified across protocols. A tree of possible records is not automatically a tree of physical worlds. WR-V therefore formalizes the objects on which these labels act before comparing them.

WR-I owns response equivalence and identifiable quantities [1]; WR-II owns the finite/global realizability distinction [2]; WR-III owns response-fibre geometry [3]; WR-IV owns the additional history-projection layer and its distinction between compatibility and probabilistic conditioning [4]. The new obligation here is to separate an outcome marginal from a declared experimental transition and to state which extra observations identify that transition. Re-proving a generic quotient, gluing or smoothing theorem would not meet this obligation.

All main constructions are finite and classical. Probability is supplied as model structure. The contribution is a typed interface and an auditable transition-completion problem, not discovery of stochastic instruments, null-space criteria or contextuality. No result identifies Bayesian concentration with physical collapse, support multiplicity with physical branching, or a failed global-variable model with absence of reality. A label of a local state must not silently become a complete cosmological world.

2 Global compatibility and finite experimental labels

Start with a nonempty globally compatible class $\mathfrak{C}(D)$ as in WR-II–IV. Declare a finite set X of input labels and a total map $\rho_- : \mathfrak{C}(D) \rightarrow X$, with $X = \rho_-(\mathfrak{C}(D))$. A label can be a finite hypothesis or a projected state at a supplied temporal cut. It is not necessarily a full realization. Declare a finite nonempty set S of successor labels, a finite protocol set E and finite nonempty outcome sets R_e . These carriers and all identifications are fixed within an experiment.

Definition 2.1 (Finite stochastic instrument). *For each $e \in E$, an instrument is an array $I_e(r, s \mid x)$ satisfying*

$$I_e(r, s \mid x) \geq 0, \quad \sum_{r \in R_e} \sum_{s \in S} I_e(r, s \mid x) = 1 \quad (x \in X). \quad (1)$$

Its outcome likelihood and conditional successor distribution are

$$L_e(r \mid x) = \sum_s I_e(r, s \mid x), \quad K_e(s \mid x, r) = \frac{I_e(r, s \mid x)}{L_e(r \mid x)} \quad \text{if } L_e(r \mid x) > 0. \quad (2)$$

When $L_e(r \mid x) = 0$, the entire corresponding instrument row is zero; a conditional K_e there has no observational meaning and is not an identifiable parameter. The outcome-plus-state distinction is classical measurement theory; the general instrument concept predates WREH [5]. This finite classical definition is not asserted to capture every quantum instrument.

The support relation is $\Gamma_e = \{(x, r, s) : I_e(r, s \mid x) > 0\}$. It can also be declared directly without probabilities; the stochastic model then refines it. With prior $p(x) > 0$ on X , an outcome of positive probability $Z_e(r) = \sum_x p(x) L_e(r \mid x)$ gives two different laws:

$$p_e^-(x \mid r) = \frac{p(x) L_e(r \mid x)}{Z_e(r)}, \quad p_e^+(s \mid r) = \frac{\sum_x p(x) I_e(r, s \mid x)}{Z_e(r)}. \quad (3)$$

The first is posterior uncertainty about an input label; the second describes the successor label in the supplied instrument model. Their domains can differ. A prior with zeros would replace X by its support in these probability statements; it would not prove model-level impossibility of omitted labels.

Definition 2.2 (Two compatibility images).

$$X_e(r) = \{x : L_e(r \mid x) > 0\}, \quad S_e(r) = \{s : \exists x \in X, I_e(r, s \mid x) > 0\}. \quad (4)$$

Input selection means $|X_e(r)| = 1$. Successor selection means $|S_e(r)| = 1$. Selection of a projected label is not selection of a unique full realization unless the relevant projection is injective on the surviving global class.

3 Four scenario labels and their different domains

The programme labels become questions about specified objects, as follows. They are not four disjoint events. CONTEXTUALIZE addresses a model's cross-context identifications, whereas SELECT and REMAIN A CLASS describe cardinality relative to an identified carrier. BRANCH requires specifying which mathematical branching is meant, and physical branching needs an additional physical model.

Label	Declared mathematical object	What is not implied
BRANCH	Outcome tree, or more than one successor in $\text{supp } K_e(\cdot \mid x, r)$.	Multiple physically coexisting worlds.
CONTEXTUALIZE	Context-indexed variables with stated overlap identifications.	Automatic absence of a global model, or a quantum implementation.
SELECT	A singleton input, successor, global or instrument-completion class; specify which.	Ontology inferred from posterior concentration.
REMAIN A CLASS	More than one candidate remains in that same specified class.	Permanent impossibility of every future distinction.

Example 3.1 (Local-state change does not violate global refinement). Let the input class be $\{0\}$, the successor carrier be $S = \{0, 1\}$ and the instrument have one outcome with $I(r, 1 \mid 0) = 1$. The successor class is $\{1\}$; it is not a subset of the input class. However, in the fixed global trajectory carrier $\Omega_e \subseteq X \times R_e \times S$, observation still selects the subset $\{\omega : r(\omega) = r\}$. The two statements concern different carriers. The nested-set result of WR-IV is preserved for full trajectories; it was never a claim that projected local states cannot change.

A stochastic successor law can represent unresolved dynamics or unrecorded classical variables. Calling its support “branches” is a model convention, not evidence that all alternatives physically occur. A deterministic transition can preserve an ambiguous input class, while a unique input can have a non-singleton successor support. Thus even within one context, selection and successor multiplicity can coexist.

4 What outcome likelihoods leave undetermined

Theorem 4.1 (Outcome-marginal completion class). *Fix normalized likelihoods $L_e(r \mid x)$ and impose no further constraints on the successor distributions. The class of instruments with these likelihoods is affinely isomorphic to*

$$\mathfrak{I}(L) \cong \prod_{(e,x,r): L_e(r|x)>0} \Delta(S), \quad \dim \mathfrak{I}(L) = N_+(|S| - 1), \quad (5)$$

where N_+ counts the positive likelihood rows. In particular, knowing all these outcome likelihoods need not identify a successor distribution, even when the input label is known.

Proof. For each positive row choose any $k_{e,x,r} \in \Delta(S)$ and define $I_e(r, s \mid x) = L_e(r \mid x)k_{e,x,r}(s)$. Set every zero-likelihood row to zero. This gives every admissible instrument once, by (2). Each positive-row scaling is a bijective affine map of a simplex. The product has the displayed dimension. Coupling, conservation or dynamical restrictions would change this class and are excluded from the statement. $\square$

Example 4.2 (The same outcomes, different successors). Take $X = \{x_0\}$, $R = \{0, 1\}$, $S = \{s_0, s_1, s_2\}$ and $L(r \mid x_0) = 1/2$. For both outcomes, choose respectively $K^A = \delta_{s_0}$, $K^B = \delta_{s_1}$ or $K^C = (1/2, 0, 1/2)$. All three instruments give exactly the same outcome distribution. Input selection already holds, but successor supports are respectively $\{s_0\}$, $\{s_1\}$ and $\{s_0, s_2\}$. No statistic of those immediate outcomes can distinguish these declared instruments. A calibrated later probe can, if its response differs on the relevant successors.

This is a deliberately strong non-identification statement: it grants exact knowledge of likelihoods for every calibrated input. In an actual unknown-input experiment, only prior-averaged or otherwise accessible responses may be known, and the ambiguity can be larger. The theorem is an elementary marginalization result, not a new impossibility theorem for all future physical experiments.

5 Successor probes and transition-completion certificates

Fix one positive row (e, x, r) and write $k(s) = K_e(s \mid x, r)$, with $n = |S|$. A calibrated probe j has a selected recorded event whose probability on successor s is $a_j(s) \in [0, 1]$. The conditional event frequency in ideal repeated preparations is $m_j = \sum_s a_j(s)k(s)$. The model assumes that this response depends only on s and that the calibration remains valid after the first experiment. A hidden dependence on x, r or unrepresented degrees of freedom must be included in the carrier or calibration.

Different probes are measured on separate repeat preparations. We do not assume joint measurability or nondisturbance of several probes on one run. The exact m_j below are ideal model-level probabilities; finite counts provide uncertainty regions, not exact equalities.

Definition 5.1 (Probe completion fibre).

$$B = \begin{pmatrix} \mathbf{1}^T \\ a_1^T \\ \vdots \\ a_q^T \end{pmatrix}, \quad b = (1, m_1, \dots, m_q)^T, \quad \mathfrak{F}(B, b) = \{k \in \mathbb{R}^n : k \geq 0, Bk = b\}. \quad (6)$$

Assume this fibre is nonempty before discussing its uniqueness. It is a compact convex subset of $\Delta(S)$. Instrument selection means that every relevant positive row is uniquely determined, with any cross-row constraints also enforced.

Write $\mathfrak{I}(L, m)$ for the subset of $\mathfrak{I}(L)$ satisfying all declared calibrated positive-row probe constraints. Rows with no probes retain their unconstrained conditional laws.

Theorem 5.2 (Uniform identification). *The map $k \mapsto Bk$ is injective on $\Delta(S)$ if and only if $\text{rank } B = n$. If the rank is deficient, every strictly positive $k \in \Delta(S)$ belongs to a non-singleton fibre.*

Proof. Full column rank gives $B(k - k') = 0 \Rightarrow k = k'$. Conversely, choose $0 \neq v \in \ker B$. The first row gives $\mathbf{1}^T v = 0$. For strictly positive k , both $k + \varepsilon v$ and $k - \varepsilon v$ are nonnegative for sufficiently small $\varepsilon > 0$, remain normalized and have the same image. They are distinct. Thus failure of full rank defeats uniform injectivity and also uniqueness at every interior point. $\square$

Theorem 5.3 (Pointwise uniqueness at a support boundary). *For a feasible k , let $Z(k) = \{s : k(s) = 0\}$ and $T_k = \{v \in \mathbb{R}^n : v(s) \geq 0 \text{ for } s \in Z(k)\}$. Then*

$$\mathfrak{F}(B, Bk) = \{k\} \iff \ker B \cap T_k = \{0\}. \quad (7)$$

Proof. A distinct feasible k' gives $v = k' - k \neq 0$, $Bv = 0$, and $v(s) = k'(s) \geq 0$ wherever $k(s) = 0$. Conversely, suppose $0 \neq v \in \ker B \cap T_k$. For coordinates with $k(s) = 0$, $k(s) + \varepsilon v(s) \geq 0$ for every $\varepsilon \geq 0$. At each remaining coordinate choose $\varepsilon > 0$ sufficiently small to preserve nonnegativity; finiteness permits one common choice. Then $k + \varepsilon v$ is normalized because $\mathbf{1}^T v = 0$, is distinct from k and has the same response. $\square$

The normalization constraint is included in $\ker B$; T_k records only the one-sided nonnegativity conditions. This is a standard feasible-direction criterion. Nonnegative uniqueness at deficient rank is established prior art [6]; no priority for (7) is claimed. Merely testing linear independence of columns on the positive support is insufficient: another feasible point can introduce new nonzero coordinates outside that support.

Proposition 5.4 (Identifiable linear quantities). *For $c \in \mathbb{R}^n$, the quantity $c^T k$ is constant on every nonempty probe fibre if and only if $c \in \text{row } B$.*

Proof. If $c^T = u^T B$, then $c^T k = u^T b$ on each fibre. If $c \notin \text{row } B$, orthogonal decomposition gives some $v \in \ker B$ with $c^T v \neq 0$. Perturb a strictly positive normalized k by $\pm \varepsilon v$ as above. The two feasible points have equal responses but different $c^T k$. $\square$

This is a uniform criterion. A particular singleton or boundary fibre can identify additional quantities. An identifiable mean does not imply an identifiable successor law. This is the WR-I identifiable-quantity distinction applied to a new transition carrier, not a replacement theorem for WR-I.

6 A three-state experiment and finite resolution

Let $S = \{s_0, s_1, s_2\}$ and let a calibrated binary detector have click probabilities $a = (0, 1/2, 1)$. For one fixed conditional instrument row, the exact click probability m gives

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1/2 & 1 \end{pmatrix}, \quad (8)$$

$$k(t) = (1 - 2m + t, 2m - 2t, t),$$

$$\max(0, 2m - 1) \leq t \leq m, \quad 0 \leq m \leq 1.$$

The formula follows by setting $t = k_2$, solving the two equations and imposing nonnegativity. Data outside $[0, 1]$ have an empty fibre. At $m = 0$ and $m = 1$ the interval collapses and the successor distribution is unique despite rank two. At every $0 < m < 1$ the interval has positive length.

Example 6.1 (A deterministic candidate is not a certificate). At $m = 1/2$, the entire fibre is $(t, 1 - 2t, t)$, $0 \leq t \leq 1/2$. It includes both the deterministic middle state $(0, 1, 0)$ and the equally weighted endpoint mixture $(1/2, 0, 1/2)$. The three columns of B are distinct, so exact responses distinguish the three pure states if purity is imposed in advance. They do not distinguish all mixtures. Choosing the deterministic candidate by a simplicity rule adds a selection convention; the data alone do not certify it.

A second independently calibrated detector with click probabilities $d = (0, 0, 1)$ measures $n_2 = k_2$. The combined matrix and inverse are

$$\tilde{B} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1/2 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \det \tilde{B} = 1/2, \quad k = (1 - 2m + n_2, 2m - 2n_2, n_2). \quad (9)$$

Feasibility is exactly $0 \leq m \leq 1$ and $\max(0, 2m - 1) \leq n_2 \leq m$. Thus two calibrated probabilities select every feasible successor law in this finite model. They still do not select a microscopic lift of the labels or a complete world without an injective bridge.

6.1 Exact uniqueness and uncertainty regions

At finite resolution replace an equality by a declared error bound:

$$\mathfrak{F}_\varepsilon(m) = \{k \in \Delta(S) : |a^T k - m| \leq \varepsilon\}. \quad (10)$$

For every $\varepsilon > 0$, the exact boundary certificate at $m = 0$ loses uniqueness: both $(1, 0, 0)$ and $(1 - \delta, \delta, 0)$ are feasible for $0 < \delta \leq \min(1, 2\varepsilon)$. An analogous perturbation applies at $m = 1$. This is not a false exact theorem; it is a change of data semantics.

With two probes, if measured values $\hat{m}, \hat{n}_2$ satisfy $|m - \hat{m}| \leq \eta_m$ and $|n_2 - \hat{n}_2| \leq \eta_n$, the affine reconstruction $\hat{k}$ from (9) obeys

$$|k_0 - \hat{k}_0| \leq 2\eta_m + \eta_n, \quad |k_1 - \hat{k}_1| \leq 2\eta_m + 2\eta_n, \quad |k_2 - \hat{k}_2| \leq \eta_n. \quad (11)$$

The reconstructed $\hat{k}$ need not itself be feasible; one should retain the nonnegative uncertainty fibre instead of silently clipping coordinates. Calibration error in the detector matrices needs its own bounds and is not included in (11). Finite counts also need a specified statistical coverage statement. No experiment has been performed for this benchmark.

7 Contextualization changes an identification obligation

Indexing a response by a protocol does not by itself establish contextuality in the sense of a no-global-model result. A contextuality test must declare variables, contexts, outcome sets, overlap identifications and an empirical model. This distinction is explicit in the global-section approach of Abramsky and Brandenburger [7]. The following abstract example uses that established language only to test the meaning of CONTEXTUALIZE; it is not a new WR-II existence theorem.

Example 7.1 (An abstract anti-correlation triangle). Let A, B, C be binary labels and the measured contexts be AB, BC, AC . In each context put probability $1/2$ on each of the two opposite-bit outcomes and zero on equal bits. Every single-variable marginal is uniform, so the overlaps agree. A context-independent global assignment would have to satisfy

$$A \oplus B = 1, \quad B \oplus C = 1, \quad A \oplus C = 1. \quad (12)$$

XOR of the three equations gives $0 = 1$, so there is no such assignment. Nor is there a probability law on global assignments with the given marginals: each constraint would hold almost surely, hence their finite intersection would hold almost surely, although it is empty.

Replacing A by separate A_{AB}, A_{AC} , and similarly for B, C , removes the overlap identities. Each of the three disconnected pairs can then choose either opposite-bit assignment; their product has eight global assignments and a product probability law. This is a different variable model, not a solution of the original identified-variable model. Contextualization can therefore be a declared relaxation of identity constraints. It does not follow that an experiment creates context-specific reality.

The triangle is an abstract overlap-compatible empirical model. No physical, quantum or nondisturbing implementation is asserted. If the real apparatus perturbs a variable between contexts, the overlap identities themselves need an independent operational justification. Selection of one context's outcome can coexist with failure of global extension across all contexts. Thus SELECT and CONTEXTUALIZE do not form mutually exclusive ontological alternatives.

8 Sequential evidence and exact selection

For this section only, let X be a persistent finite latent-hypothesis set with full-support prior p_0 . It is not a changing state carrier. An adaptive policy chooses e_t from the observed past record h_{t-1} . The model supplies the likelihood $L_{e_t}(r_t | x, h_{t-1})$, including any history dependence. The policy cannot access the unknown x except through the record.

Proposition 8.1 (Positive likelihood blocks finite hard selection). *For a finite realized record h_N , assume each realized conditional likelihood is positive for every $x \in X$. Then the posterior has full support on X and the hard compatibility class remains X . In particular, if $|X| > 1$, no such record selects a unique latent hypothesis.*

Proof. The policy is fixed by the record. Bayes' formula gives

$$p_N(x \mid h_N) = \frac{p_0(x) \prod_{t=1}^N L_{e_t}(r_t \mid x, h_{t-1})}{\sum_{x'} p_0(x') \prod_{t=1}^N L_{e_t}(r_t \mid x', h_{t-1})} > 0. \quad (13)$$

Every hypothesis has positive likelihood, so none is removed by the finite record. This argument requires neither independence of outcomes nor a nonadaptive protocol sequence. $\square$

For example, two equally weighted persistent hypotheses can have independent binary outcomes with probabilities of one equal to 0.9 and 0.1. After three ones, the first posterior probability is $0.9^3 / (0.9^3 + 0.1^3) = 729/730$, while both hypotheses remain compatible. A confidence threshold or MAP estimate chooses an action or estimator; it is not a proof that the remaining compatibility set is a singleton. The statement concerns finite records; it does not preclude asymptotic identification under additional assumptions.

Comparison of statistical experiments by their decision value is classical [8]. A protocol that improves a chosen risk is not thereby a protocol that selects a unique realization. Conversely, an exact deterministic response can isolate a label without identifying every hidden microscopic representative. The decision criterion, compatibility criterion and transition criterion must be stated separately.

9 Provenance, failure criteria and downstream handoff

Theorems 4.1–5.3 are self-contained elementary results about marginalization, kernels and nonnegative linear systems. Proposition 5.4 is row-space factorization. Proposition 8.1 is positivity under finite Bayesian conditioning. They are not presented as discoveries of new mathematical principles. Davies–Lewis provides the pre-existing outcome/state instrument distinction; Donoho–Tanner supplies adjacent nonnegative inverse-problem prior art; Abramsky–Brandenburger supplies the global-section contextuality language; Blackwell supplies the classical comparison-of-experiments setting. None is claimed to have established the entire WREH architecture.

The new carrier is the class of admissible experimental transition descriptions. Once declared, its response fibres can use the inherited WR-I–III language. WR-V does not reopen generic fibre geometry or claim that a simplex boundary is a physical boundary of the Universe. Its negative result is acceptable: immediate outcome data need not determine a successor model. Its positive certificate is deliberately conditional: calibrated post-experiment responses and stated admissibility assumptions can determine a finite transition law.

A stronger research claim would require a problem-specific new result and a tailored literature audit. In particular, operational calibration, finite-sample confidence, restrictions coupling instrument rows, inaccessible inputs and quantum instruments remain separate obligations. This draft neither certifies exhaustive originality nor claims publication readiness. Its first benchmark is analytically reproducible and is supplied with exact rational checks; computation does not replace the proofs.

For WR-VI the interface is a declared physical admissibility subset $\mathcal{A}_{\text{phys}}$ of instrument completions. One may then ask whether $\mathcal{I}(L, m) \cap \mathcal{A}_{\text{phys}}$ is empty, unique or nonunique. Conservation, causal, energetic or thermodynamic conditions must be independently specified and justified. Empty intersection means failure of the declared model-and-data combination; it is not automatically a wall of physical existence. For WR-IX, indistinguishability is always relative to the admissible protocol family and resource regime; an untested future probe can invalidate an outcome-only equivalence.

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